Quadratic Equations — MCQ Worksheet

Mathematics • Chapter 4 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. A quadratic equation in x has the form:

(A) $ax + b = 0$
(B) $ax^2 + bx + c = 0, a \neq 0$
(C) $ax^3 + bx^2 + cx + d = 0$
(D) $ax + by + c = 0$

Q2. The roots of a quadratic are also called:

(A) Coefficients
(B) Constants
(C) Zeros / solutions
(D) Variables

Q3. The standard formula for roots of $ax^2 + bx + c = 0$ is x =

(A) $[-b \pm \sqrt{(b^2 - 4ac)}]/(2a)$
(B) $b^2 - 4ac$
(C) $[b \pm \sqrt{(b^2 - 4ac)}]/(2a)$
(D) $[-b \pm \sqrt{(b^2 + 4ac)}]/(2a)$

Q4. The discriminant is:

(A) $b^2 - 4ac$
(B) $b^2 + 4ac$
(C) 2a
(D) b/a

Q5. Roots are real and distinct if D:

(A) = 0
(B) > 0
(C) < 0
(D) = b

Q6. Roots are real and equal if D:

(A) > 0
(B) = 0
(C) < 0
(D) = 1

Q7. Roots are not real (complex) if D:

(A) > 0
(B) = 0
(C) < 0
(D) Negative integer only

Q8. Roots of $x^2 - 5x + 6 = 0$ are:

(A) 2, 3
(B) –2, –3
(C) 1, 6
(D) 2, –3

Q9. Roots of $x^2 - 4 = 0$ are:

(A) 2 only
(B) 2 and –2
(C) ±4
(D) $\sqrt{2}$ and $-\sqrt{2}$

Q10. Roots of $2x^2 + x - 3 = 0$ are:

(A) 1, –3/2
(B) –1, 3/2
(C) 1/2, –3
(D) –1/2, 3

Q11. Discriminant of $x^2 - 4x + 4 = 0$ is:

(A) 0
(B) 4
(C) 16
(D) –16

Q12. Discriminant of $3x^2 + 5x + 1 = 0$ is:

(A) 13
(B) 17
(C) 25
(D) 9

Q13. The equation $x^2 + 2x + 5 = 0$ has:

(A) Two real roots
(B) Two equal roots
(C) No real roots
(D) One real root

Q14. For $x^2 + kx + 4 = 0$ to have equal roots, k =

(A) ±2
(B) ±4
(C) ±6
(D) ±8

Q15. For $2x^2 - 6x + p = 0$ to have equal roots, p =

(A) 3/2
(B) 9/2
(C) 18
(D) 9

Q16. Sum of roots of $2x^2 - 4x - 6 = 0$ is:

(A) –2
(B) 2
(C) 3
(D) –3

Q17. Product of roots of $2x^2 - 4x - 6 = 0$ is:

(A) –3
(B) 3
(C) 2
(D) –6

Q18. Quadratic with roots 2 and 5 is:

(A) $x^2 - 7x + 10 = 0$
(B) $x^2 + 7x + 10 = 0$
(C) $x^2 - 3x + 10 = 0$
(D) $x^2 - 7x - 10 = 0$

Q19. If one root of $x^2 + ax + 6 = 0$ is 2, then a =

(A) –5
(B) 5
(C) –4
(D) 4

Q20. Method of completing the square is used to:

(A) Simplify equation
(B) Solve quadratic
(C) Add fractions
(D) Subtract polynomials

Q21. A quadratic obtained from problem of speed: 'A train covers 360 km. If speed had been 5 km/h more, time would be 1 h less.' The speed is:

(A) 40 km/h
(B) 45 km/h
(C) 60 km/h
(D) 30 km/h

Q22. Sum of two numbers is 27 and product is 182. The numbers are:

(A) 13, 14
(B) 12, 15
(C) 11, 16
(D) 10, 17

Q23. The number of real solutions of $x^2 + 1/x^2 = 2$ is (with $x \neq 0$):

(A) 0
(B) 1
(C) 2
(D) Infinitely many

Q24. A quadratic equation with real coefficients always has at most ___ real roots:

(A) 1
(B) 2
(C) 3
(D) Infinite

Q25. If sum of roots of a quadratic is 6 and product is 9, the equation is:

(A) $x^2 - 6x + 9 = 0$
(B) $x^2 + 6x + 9 = 0$
(C) $x^2 - 6x - 9 = 0$
(D) $x^2 + 9x + 6 = 0$

— End of Worksheet —